

Unit - I basics

1. Write a program to print values of variables of different data types.
2. Write a program to read and print values of variables of different data types.
3. Write a program to calculate the area of a triangle using herons formulae.
($area = \sqrt{S(S-a)(S-b)(S-c)}$)).
4. Write a program to calculate the distance between two points.
5. Write a program to perform addition ,subtraction, multiplication division and modulo division on two integer numbers.
6. Write a C++ program to perform addition ,subtraction, multiplication division and modulo division on two floating point numbers.
7. Write a C++ program that demonstrates the use of relational operators.
8. Write a C++ program to illustrate the use of unary prefix increment and decrement operators.
9. Write a C++ program to illustrate the use of unary postfix increment and decrement operators.
10. Write a C++ program to find the largest of two numbers using the ternary operator.
11. Write a C++ program to find the largest of three numbers using the ternary operator.
12. Write a C++ program to demonstrate the use of assignment operators.
13. Write a C++ program to calculate the area of a circle.
14. Write a C++ program to read a character in uppercase and then print it in lowercase.
15. write a C++ program to print the digit at one's place of a number.
16. Write a C++ program to swap two numbers without using a temporary variable.
17. Write a C++ program to swap two numbers with a temporary variable.
18. Write a C++ program to calculate the average of two numbers. Print their deviation.
19. Write a C++ program to convert degrees fahrenheit into degrees celsius.
20. Write a C++ program to calculate the total amount of money in the piggy bank given the coins of ₹10, ₹5, ₹2, ₹1.

21. Write a C++ program to swap the first and last digits of any number.
22. Write a C++ program that replaces all the words of "string1" with "string2".
23. Write a C++ program to compute the sum of two given integer values. If the two values are the same, then return triple their sum.
24. Write a C++ program to remove the character at a given position in the string.
The given position will be in the range 0..n string length -1 inclusive.
25. Write a C++ program to exchange the first and last characters in a given string and return the result string.
26. Write a C++ program to check which number is closest to the value 100 among two given integers. Return 0 if the two numbers are equal.
27. Write a C++ program to check whether two given integers are in the range [1,...,n] inclusive, or they are both in the range [1,...,n] inclusive.
28. Write a C++ program to check if two given non-negative integers have the same last digit.
29. Write a C++ program to create another string which is n copies of the first 3 characters of a given string. If the length of the given string is less than 3 then return n copies of the string.
30. Write a C++ program to find the largest value from the first, last, and middle elements of a given array of integers of odd length (at least 1).

Unit - II basics

1. Write a program to check and increment a number if it is positive.
2. Write a program to determine whether a person is eligible to vote.
3. write a program to determine the character entered by the user.
4. Write a program to find the larger number between two numbers.
5. write a program to find whether the given number is even or odd.
6. Write a program to enter any character. If the entered character is in lowercase, convert it into uppercase. If it is a lower case character, then convert it into uppercase.
7. A company decides to give bonuses to all its employees on diwali. A five percent bonus on salary is given to male workers and a 10% bonus on salary to female workers. write a program to enter the salary and gender of the employee. If the salary is less than ₹10000 then the employee gets an extra two per cent bonus on salary. Calculate the bonus that has to be given to the employees and display the salary that the employee will get.
8. Write a program to find whether the given year is a leap year or not.
9. Write a program to find the greatest number from three numbers.
10. Write a program to take an input from the user and then check whether it is a number or a character. If it is a character, determine whether it is upper case or lower case.
11. Write a program to calculate roots of a quadratic equation.
12. Write a program to determine whether an entered character is a vowel or not.
13. Write a program that accepts a number from 1-10. Print whether the number is even or odd using a switch-case construct.
14. Write a program to enter a number from 1-7 and display the corresponding day of the week using switch-case statements.
15. Write a program to calculate the sum of the first 10 numbers.
16. Write a program to print 20 horizontal asterisks(*).
17. Write a program to calculate the sum of numbers from m to n.
18. Write a program to read the numbers until -1 is encountered. count the negative positives and zeros entered by the user.

19. Write a program to calculate the average of numbers entered by the user.
20. Write a program using do-while loop to display the square and cube of first n natural numbers.
21. Write a program to list all the leap years from 1900 to 2100.
22. Write a program to print the following patterns.

12345	1
12345	22
12345	333
12345	4444
12345	55555
1	1
21	121
321	12321
4321	1234321
54321	123454321

23. Write a program to print the multiplication table of n, where n is entered by the user.
24. Write a program to print all the numbers from m-n thereby classifying them as even or odd.
25. Write a program to calculate the sum of all positive numbers entered by the user by using goto statement.

Unit - III functions

1. To add two integers using functions.
2. Write a program to find the biggest of three integers using functions.
3. Write a program to convert time into minutes using functions.
4. Write a program to calculate simple interest and compound interest using a function.
5. Write a program to sum the series $1/1! + 1/2! + 1/3! + \dots + 1/n!$ using a function.
6. Write a program to calculate the volume of a cuboid using default arguments.
7. Write a program to calculate simple interest. Suppose the customer is a senior citizen. He is being offered 12 per cent rate of interest for all other customers the roi is 10 percent.
8. Write a program to print the following pattern using default arguments.
9. Write a program to display values of different data types using compile time polymorphism.
10. Write a program to add two values of different data types using static polymorphism.
11. Write a program that finds the absolute value of a number.
12. Write a program to print the fibonacci series without using recursion.
13. Write a program using functions to perform calculator operations.
14. Write a function that converts temperature given from celsius into fahrenheit.
15. Write a program to find whether a given number is even or odd using functions.
16. Write a program to reverse a string using recursion.
17. Write a program to calculate $\exp(x,y)$ using recursion.
18. Write a program to swap two integers using a call by value method of passing arguments to a function.
19. Write a program to swap two integers using a call by reference method of passing arguments to a function.
20. Write a program to swap two integers using a call by value method of passing arguments to a function.

Unit - IV arrays

1. Write a program to read and display n numbers using an array
2. Write a program to read and display n random numbers using an array
3. Write a program to find the largest of n numbers using arrays.
4. Write a program to interchange the largest and the smallest number in the array.
5. Write a program to find the second largest number using an array of n numbers.
6. Write a program to enter n number of digits and then form a number using these digits.
7. Write a program to find whether the array of integers contain a duplicate number.
8. Write a program to read marks of 50 students in the range of 0 - 100. Then make 10 groups such as 0-10,10-20,20-30, and so on. Count the number of values that fall into each group and display the result.
9. Modify the previous program to display frequency histograms of each group.
10. Write a program to read a sorted list of floating point values and then calculate and display the median of the values.
11. Write a program to insert a number at a given location in an array.
12. Write a program to insert a number in an array that is already sorted in ascending order.
13. Write a program to merge two unsorted arrays.
14. Write a program to merge two sorted arrays.
15. Write a program to implement linear search in any given array.
16. Write a program to implement binary search.
17. Write a program to read an array of n numbers and then find out the smallest number.
18. Write a program to merge two integer arrays. Display the merged array in reverse order.
19. Write a program to interchange the biggest and smallest number in the array using functions.
20. Write a program to print the elements of a 2D array.
21. Write a program to generate the pascals triangle.
22. In a small company there are five salesmen. Each salesman is supposed to sell 3 products. Write a program using two dimensional array to print
 - a. The total sales by each salesman and
 - b. Total sales of each item.
23. In a class there are 10 students. Each student is supposed to appear in three tests. write a program using a two dimensional array to print the marks obtained by each student in different subjects, total marks and average obtained by each student and store the average of each student in a separate 1D array so that it can be used to calculate the class average.

24. Write a program to read the marks in a two dimensional array which stores marks of 10 students in five subjects. Write a program to display the highest marks in each subject.
25. Write a program to transpose a 3 x 3 matrix.
26. Write a program to multiply two m x n matrices.
27. Write a menu driven program to read and display an mxn matrix. Find the sum transpose and product of two mxn matrices.
28. Write a program to fill a square matrix with values zero on the diagonals, 1 on the upper right triangle and -1 on the lower left triangle.
29. Write a program to read and display a 2x2x2 array.

Unit - V Strings

1. Write a program to find the length of a string.
2. Write a program to convert characters of a string in upper case.
3. Write a program to convert characters of a string into lowercase.
4. Write a program to concatenate two strings.
5. Write a program to append a string to another string.
6. Write a program to compare two strings.
7. Write a program to reverse the given string.
8. Write a program to swap two strings.
9. Write a program to extract the substring from a given string.
10. Write a program to insert a string in the main text.
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